

ECE 6360 Parallel and Heterogeneous Computing

Homework 4: Multithreading

State where each allocated item in the following code snippet is stored (stack or heap) and compute the total number of bytes for each:

```
int a = 12;
float x[9];
double y* = (double*) malloc(sizeof(double) * 10);
```

Describe the errors that can result from the following writes:

```
int N = 2000;
double* x = (double*) malloc(N * sizeof(double));
double y[5];
double z[7];
x[2000] = 3.14159;
y[5] = 2.71828;
z[7] = 6.67430;
```

Declare a function pointer `op` that can point to any function taking two `float` arguments that returns an `int`:

Write a code snippet that launches 10 threads calling this function for each element in the following arrays:

```
float x[12];
float y[12];
int a[12];
//...assume x and y are filled here, write your code below
```

Suppose this code reads and writes to `x`, `y`, and `a` right after the launch loop. What errors can be caused in each case and how do you prevent them?